

Paper Trading System Architecture

This document outlines the modular approach to building a paper trading system for the IDX market, separating fundamental analysis (the "What") from technical execution (the "When").

1. Architectural Overview

The system is designed as a pipeline with distinct responsibilities to ensure clean separation of concerns.

Layer	Function
Data Ingestion	Fetch raw market and financial data.
Fundamental Engine	"What" – Filter for accumulation and growth.
Technical Engine	"When" – Monitor triggers for entry.
Paper Trade Logic	"Execution" – Manage virtual P&L.

2. Implementation Modules

Fundamental Engine ("The What")

This engine runs on a schedule (e.g., end-of-day). It filters the market to create a "Universe" of stocks worth monitoring.

- **Quantitative Screening:** Filter by financial health metrics (ROE, PE, PB).
- **Institutional Tracking:** Analyze foreign net flows (Smart Money).

Technical Engine ("The When")

This engine runs in real-time or on bar-close to check if "What" stocks hit a buying trigger.

- **Indicators:** Use pandas-ta or TA-Lib.
- **Trigger Logic:** Detect specific chart patterns (e.g., breakout, MA crossover).

3. Suggested Tech Stack

- **Language:** Python
- **Libraries:** pandas, numpy, ta-lib
- **Storage:** SQLite / PostgreSQL for historical data and state.
- **Orchestration:** Cron jobs or Airflow for scheduling.

Generated for system development reference.